Exploring Trends in the Automotive Industry Using SQL

An End-to-End SQL Portfolio Project

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Role:

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Tools Used:

SQL (MySQL), MySQL Workbench

Project Type:

Data Analytics Portfolio Project

Year:

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Project Overview :-

This project focuses on analyzing trends in the automotive industry using SQL. The dataset contains information on more than 8,000 vehicles, including their selling prices, manufacturing years, fuel types, ownership history, transmission systems, mileage, engine specifications, and seating capacity.

The automotive industry is highly influenced by factors such as vehicle age, fuel efficiency, ownership history, and technological advancements. Understanding these factors is essential for manufacturers, dealerships, and consumers to make informed decisions regarding vehicle pricing, market demand, and resale value.

The primary objective of this project is to apply SQL techniques in a real-world business scenario to uncover valuable insights about the used-car market. Through descriptive analysis, aggregations, subqueries, Common Table Expressions (CTEs), and Window Functions, the project investigates pricing patterns, customer preferences, fuel trends, vehicle performance, and ownership impacts.

By transforming raw automotive data into meaningful insights, this project demonstrates how SQL can be used to support data-driven decision-making within the automotive sector. The findings can help dealerships optimize inventory, assist buyers in identifying value-for-money vehicles, and provide market intelligence for future business strategies.

Dataset Description :-

The dataset used in this project contains vehicle listing records collected from the automotive market. Each row represents a single vehicle and includes details related to pricing, performance, fuel efficiency, ownership history, and technical specifications.

DATASET DETAILS:

- Dataset Name: Exploring Trends in the Automotive Industry
- Total Records: 8,148
- Total Columns: 13
- Vehicle Types: Multiple Car Brands and Models
- Fuel Categories: Petrol, Diesel, CNG, LPG, Electric
- Transmission Types: Manual, Automatic
- Ownership Categories: First Owner, Second Owner, Third Owner, Test Drive Car

Attribute	Description
Name	Car Model
Year	Manufacturing Year
Selling Price	Vehicle Selling Price
Fuel	Fuel Type
Transmission	Manual / Automatic
Mileage	Fuel Efficiency
Engine	Engine Capacity
Seats	Seating Capacity
Owner	Ownership Category

Dataset Summary

- Total Records: 8148
- Total Columns: 13
- Multiple Fuel Types
- Manual & Automatic Vehicles
- Various Ownership Categories

Exploring Trends in the Automotive Industry

REPORT | SUMMARY

<u>Basic Dataset Overview & Descriptive Analysis :-</u>

Key Findings:

1: Total Cars

```
SELECT
    COUNT(*) AS total_cars
FROM car_info;
```

total_cars
7927

Explanation:
This query counts the total number of car records available in the dataset. Each row represents a single car listing, so the result shows the overall size of the dataset.

2 : Distinct Fuel Types

```
SELECT
    COUNT(DISTINCT fuel)
FROM car_info;
```

COUNT(DISTINCT fuel)
5

Explanation:
This query calculates the number of unique fuel types present in the dataset. It helps understand the variety of fuel options available, such as Petrol, Diesel, CNG, LPG, and Electric.

3 : Distinct Transmission Types

```
SELECT
    COUNT(DISTINCT transmission)
FROM car_info;
```

COUNT(DISTINCT transmission)
2

Explanation:
This query determines the number of different transmission types available in the dataset. It provides insights into the variety of transmission systems, such as Manual and Automatic.

4: Cars By Fuel Type

```
SELECT
    fuel, COUNT(*)
FROM car_info
GROUP BY fuel;
```

fuel	COUNT(*)
Petrol	3534
Diesel	4304
Electric	1
CNG	53
LPG	35

Explanation:
This query counts the number of cars available for each fuel type. The results help identify which fuel category is most common among the listed vehicles.

5: Average Selling Price By Fuel Type

```sql
SELECT
        fuel,
        ROUND(AVG(selling_price),2)
FROM car_info
GROUP BY fuel;
```

fuel	ROUND(AVG(selling_price),2)
Petrol	474468.17
Diesel	804352.41
Electric	2650000.00
CNG	313415.04
LPG	210885.71

Explanation:
This query calculates the average selling price for each fuel type. It helps compare the market value of vehicles based on the fuel they use.

SECTION 2: PRICE & REVENUE ANALYSIS:-

6: Highest Average Selling Price Car Models

```
SELECT
        Name,
        ROUND(AVG(selling_price),2) avg_price
FROM car_info
GROUP BY Name
ORDER BY avg_price DESC
LIMIT 10;
```

Name	avg_price
Volvo XC90 T8 Excellence BSIV	10000000.00
BMW X7 xDrive 30d DPE	7200000.00
Audi A6 35 TFSI Matrix	6223000.00
Mercedes-Benz S-Class S 350 CDI	5962500.00
BMW 6 Series GT 630d Luxury Line	5860000.00
BMW 3 Series Gran Limousine 320Ld Luxury Line	5800000.00
Volvo XC60 Inscription D5 BSIV	5500000.00
Volvo S90 D4 Inscription BSIV	5500000.00
BMW X4 M Sport X xDrive20d	5461290.32
Mercedes-Benz E-Class Exclusive E 200 BSIV	5200000.00

Explanation:
This query calculates the average selling price of each car model and ranks them in descending order. The top 10 results represent the most expensive car models on average.

7: Selling Price Trend By Year

```sql
SELECT
    year,
    ROUND(AVG(selling_price),2) avg_price
FROM car_info
GROUP BY year
ORDER BY year;
```

year	avg_price
1994	72000.00
1995	55000.00
1996	41000.00
1997	86111.00
1998	57888.89
1999	76928.57
2000	76682.13
2001	46500.00
2002	103789.42
2003	101572.92
2004	109408.92
2005	143756.57
2006	170462.24
2007	179106.55
2008	205559.78
2009	228809.48
2010	274010.63
2011	322096.45
2012	353143.28
2013	460664.62
2014	516675.74
2015	596770.27
2016	700317.72
2017	890193.04
2018	958183.60
2019	1776986.25

Explanation:
This query calculates the average selling price of cars for each manufacturing year. It helps analyze how vehicle prices vary across different production years and identify pricing trends over time.

8: Top 10 Most Expensive Cars

```
SELECT

        Name,selling_price
FROM car_info
ORDER BY selling_price DESC
LIMIT 10;
```

Name	selling_price
Volvo XC90 T8 Excellence BSIV	10000000
BMW X7 xDrive 30d DPE	7200000
Audi A6 35 TFSI Matrix	6523000
Audi A6 35 TFSI Matrix	6223000
Mercedes-Benz S-Class S 350 CDI	6000000
BMW 6 Series GT 630d Luxury Line	6000000
Mercedes-Benz S-Class S 350 CDI	6000000
Mercedes-Benz S-Class S 350 CDI	6000000
BMW 6 Series GT 630d Luxury Line	6000000
BMW 6 Series GT 630d Luxury Line	6000000

Explanation:
This query retrieves the ten cars with the highest selling prices in the dataset. It helps identify premium and luxury vehicles with the greatest market value.

SECTION 3: MILEAGE & PERFORMANCE ANALYSIS:-

9: Highest Mileage Cars

```
SELECT
            Name,
           AVG(CAST(SUBSTRING_INDEX(mileage,' ',1) AS DECIMAL(5,2))) mileage
FROM car_info
GROUP BY Name
ORDER BY mileage DESC
LIMIT 10;
```

Name	mileage
Volvo XC90 T8 Excellence BSIV	42.000000
Maruti Alto 800 CNG LXI Optional	33.440000
Maruti Alto 800 LXI CNG	33.000000
MG ZS EV Exclusive	32.520000
Mahindra XUV300 W8 Diesel Sunroof	32.520000
Nissan Magnite XV Premium	32.520000
Maruti Wagon R CNG LXI	32.520000
Maruti Alto K10 LXI CNG	32.260000
Maruti Alto 800 CNG LXI	31.453333
Maruti Swift Dzire ZDI Plus	28.400000

Explanation:
This query extracts the numerical mileage value from the mileage column, calculates the average mileage for each car model, and ranks them in descending order. The top 10 results represent the most fuel-efficient cars in the dataset.

10: Top 3 Cars Highest Average Mileage

```
SELECT
    Name,
    ROUND(AVG(CAST(SUBSTRING_INDEX(mileage,' ',1) AS DECIMAL(5,2))), 2)
AS avg_mileage
FROM car_info
GROUP BY Name
ORDER BY avg_mileage DESC
LIMIT 3;
```

Name	avg_mileage
Volvo XC90 T8 Excellence BSIV	42.00
Maruti Alto 800 CNG LXI Optional	33.44
Maruti Alto 800 LXI CNG	33.00

Explanation:
This query extracts the numeric mileage value from the mileage column and calculates the average mileage for each car model. The results are sorted in descending order of average mileage, and only the top three car models are displayed. This helps identify the most fuel-efficient vehicles in the dataset.

11: Highest Engine Capacity Cars

```sql
SELECT
        Name,
        MAX(engine)
FROM car_info
GROUP BY Name;
```

Name	MAX(engine)
Maruti Alto 800 LXI Opt	999 CC
Skoda Slavia 1.0 TSI Ambition	1956 CC
BMW 3 Series Gran Limousine 320Ld Luxury Line	998 CC
MG ZS EV Exclusive	998 CC
Tata Punch Adventure	1451 CC
Maruti S-Presso VXi Plus	999 CC
Maruti S-Presso LXi	999 CC
Hyundai Creta SX Turbo	1997 CC
Renault Kiger RXT AMT Opt DT	1956 CC
Renault KWID CLIMBER	998 CC
Mahindra XUV300 W8 Diesel Sunroof	998 CC
Mahindra XUV700 AX5 Diesel AT	1451 CC
Renault Triber RXT	1197 CC
Hyundai Creta SX Diesel AT	2993 CC
Nissan Magnite XV Premium	998 CC
Renault Triber RXZ	999 CC
Hyundai Tucson Platinum AT	1197 CC
Hyundai i20 Sportz Diesel	2198 CC
BMW X7 xDrive 30d DPE	2993 CC
Datsun GO T Option	1198 CC
Datsun RediGO 1.0 S	999 CC
Ford Freestyle Titanium Petrol BSIV	1194 CC
Honda Civic ZX Diesel BSIV	1597 CC

Explanation:
This query finds the maximum engine capacity recorded for each car model. It helps identify vehicles with larger engines, which are often associated with higher performance and power.

SECTION 4: OWNERSHIP ANALYSIS:-

12: Cars By Ownership

```
SELECT
        owner,
        COUNT(*)
FROM car_info
GROUP BY owner;
```

owner	COUNT(*)
First Owner	5232
Second Owner	2020
Test Drive Car	5
Third Owner	510
Fourth & Above Owner	160

Explanation:
This query counts the number of cars based on ownership type. It provides insights into how many vehicles are First Owner, Second Owner, Third Owner, or belong to other ownership categories.

13: Average Selling Price By Owner Type

```
SELECT
        owner,
        AVG(selling_price)
FROM car_info
GROUP BY owner;
```

owner	AVG(selling_price)
First Owner	792202.0352
Second Owner	401825.8535
Test Drive Car	4403800.0000
Third Owner	293187.2804
Fourth & Above Owner	233196.8250

Explanation:
This query calculates the average selling price for each ownership category. It helps analyze how vehicle ownership history affects resale value and market pricing.

Cost, VAT & Profitability Analysis :-

14: Most Valuable Car Models

```sql
SELECT
        Name,
        AVG(selling_price) AS avg_price
FROM car_info
GROUP BY Name
ORDER BY avg_price DESC
LIMIT 10;
```

Name	avg_price
Volvo XC90 T8 Excellence BSIV	10000000.0000
BMW X7 xDrive 30d DPE	7200000.0000
Audi A6 35 TFSI Matrix	6223000.0000
Mercedes-Benz S-Class S 350 CDI	5962500.0000
BMW 6 Series GT 630d Luxury Line	5860000.0000
BMW 3 Series Gran Limousine 320Ld Luxury Line	5800000.0000
Volvo XC60 Inscription D5 BSIV	5500000.0000
Volvo S90 D4 Inscription BSIV	5500000.0000
BMW X4 M Sport X xDrive20d	5461290.3226
Mercedes-Benz E-Class Exclusive E 200 BSIV	5200000.0000

Explanation

This query identifies the car models with the highest average selling prices, helping determine which vehicles hold the greatest market value.

15: Fuel Type With Highest Average Selling Price

```sql
SELECT
        fuel,
        ROUND(AVG(selling_price),2) avg_price
FROM car_info
GROUP BY fuel
ORDER BY avg_price DESC;
```

fuel	avg_price
Electric	2650000.00
Diesel	804352.41
Petrol	474468.17
CNG	313415.04
LPG	210885.71

Explanation
This analysis helps identify which fuel category commands the highest market prices.

16: Transmission Type Value Comparison

```sql
SELECT
        transmission,
        ROUND(AVG(selling_price),2) avg_price
FROM car_info
GROUP BY transmission;
```

transmission	avg_price
Manual	463262.73
Automatic	1887939.87

Explanation
This query compares average vehicle prices across transmission categories.

17: Ownership Impact On Vehicle Value

```sql
SELECT

        owner,
        ROUND(AVG(selling_price),2) avg_price
FROM car_info
GROUP BY owner
ORDER BY avg_price DESC;
```

owner	avg_price
Test Drive Car	4403800.00
First Owner	792202.04
Second Owner	401825.85
Third Owner	293187.28
Fourth & Above Owner	233196.83

Explanation
This query evaluates how ownership history affects resale prices.

18: Premium Vehicle Segment Analysis

```sql
SELECT

        COUNT(*)
FROM car_info
WHERE selling_price >
        (
        SELECT

                AVG(selling_price)
        FROM car_info
        );
```

COUNT(*)
2130

Explanation
This analysis measures the number of vehicles priced above the market average.

19: Most Preferred Fuel Type

```
SELECT
        fuel,
        COUNT(*) total_cars
FROM car_info
GROUP BY fuel
ORDER BY total_cars DESC;
```

fuel	total_cars
Diesel	4304
Petrol	3534
CNG	53
LPG	35
Electric	1

Explanation
This query identifies the dominant fuel category in the market.

20: Manual vs Automatic Market Share

```
SELECT
        transmission,
        COUNT(*) total_cars
FROM car_info
GROUP BY transmission;
```

transmission	total_cars
Manual	6879
Automatic	1048

Explanation
This analysis measures market preference between transmission types.

21: Most Common Ownership Category

```
SELECT
    owner,
    COUNT(*) total_cars
FROM car_info
GROUP BY owner
ORDER BY total_cars DESC;
```

owner	total_cars
First Owner	5232
Second Owner	2020
Third Owner	510
Fourth & Above Owner	160
Test Drive Car	5

Explanation
This query identifies the ownership segment that dominates the used-car market.

22: Average Mileage By Fuel Type

```
SELECT
    fuel,
    AVG(CAST(SUBSTRING_INDEX(mileage,' ',1) AS DECIMAL(10,2)))
avg_mileage
FROM car_info
GROUP BY fuel;
```

fuel	avg_mileage
Petrol	19.078016
Diesel	19.641726
Electric	32.520000
CNG	24.165660
LPG	18.577143

Explanation
This query evaluates fuel efficiency trends across fuel categories.

23: Top Brands By Number Of Listings

```
SELECT
        Name,
        COUNT(*) total_listings
FROM car_info
GROUP BY Name
ORDER BY total_listings DESC
LIMIT 10;
```

Name	total_listings
Maruti Swift Dzire VDI	162
Maruti Alto 800 LXI	82
Maruti Alto LXi	80
BMW X4 M Sport X xDrive20d	62
Maruti Swift VDI	61
Maruti Swift Dzire VXI	55
Maruti Wagon R LXI	53
Maruti Alto K10 VXI	50
Hyundai EON Era Plus	48
Maruti Ertiga VDI	45

Explanation

This query identifies the most frequently listed vehicle models in the dataset.

SECTION 5: ADVANCED SQL ANALYSIS:-

1. Calculate the average selling price for cars with a manual transmission and owned by the first owner, for each fuel type?

```
SELECT
        fuel,
        AVG(selling_price) AS avg_selling_price
FROM car_info
        WHERE transmission = 'Manual' AND owner = 'First Owner'
GROUP BY fuel;
```

fuel	avg_selling_price
Petrol	420327.3130
Diesel	640659.7283
CNG	360909.0000
LPG	250882.3529

Explanation:
This query calculates the average selling price of vehicles that have a manual transmission and are owned by their first owner. The results are grouped by fuel type, allowing comparison of resale values across different fuel categories such as Petrol, Diesel, CNG, and LPG. This helps identify which fuel type retains the highest market value among manually driven first-owner vehicles.

2. Find the top 3 car models with the highest average mileage, considering only cars with more than 5 seats?

```
SELECT
        Name,
        AVG(mileage) AS avg_mileage
FROM car_info
        WHERE seats > 5
GROUP BY Name
ORDER BY avg_mileage DESC
LIMIT 3;
```

Name	avg_mileage
Maruti Ertiga ZDI Plus	25.47
Mahindra KUV 100 D75 K8	25.320000000000004
Mahindra KUV 100 D75 K6 Plus	25.32

Explanation:
This query identifies the three car models with the highest average mileage among vehicles that have more than five seats. By focusing on larger vehicles, the analysis highlights fuel-efficient options within family and multi-passenger vehicle segments. This can help consumers identify economical vehicles with higher seating capacity.

3. Identify the car models where the difference between the maximum and minimum selling prices is greater than ₹10,000?

```
SELECT
        Name
FROM car_info
GROUP BY Name
HAVING MAX(selling_price) - MIN(selling_price) > 10000;
```

Name
Hyundai Verna 1.6 CRDi SX
Hyundai Verna 1.6 SX CRDi (O)
Hyundai Verna 1.6 VTVT AT SX
Hyundai Verna 1.6 VTVT S
Hyundai Xcent 1.2 Kappa SX Option
Jaguar XE 2016-2019 2.0L Diesel ...
Jeep Compass 2.0 Limited 4X4
Jeep Compass 2.0 Limited Option
Mahindra Bolero 2011-2019 DI - A...
Mahindra Bolero 2011-2019 EX AC
Mahindra Bolero 2011-2019 ZLX
Mahindra Bolero Power Plus Plus N...
Mahindra KUV 100 mFALCON D75 K6
Mahindra KUV 100 mFALCON D75 K8
Mahindra KUV 100 mFALCON G80 K2
Mahindra KUV 100 mFALCON G80 ...
Mahindra Scorpio 1.99 S10
Mahindra Scorpio Intelli Hybrid S6 ...
Mahindra Scorpio S2 7 Seater
Mahindra Scorpio S4 7 Seater
Mahindra Scorpio S4 Plus
Mahindra Scorpio S4 Plus 4WD
Mahindra TUV 300 T4
Mahindra TUV 300 T8
Mahindra TUV 300 T8 AMT

Explanation:
This query finds car models that exhibit significant price variation in the market. By comparing the highest and lowest selling prices for each model, it identifies vehicles whose prices differ by more than ₹10,000. Such variations may occur due to differences in manufacturing year, condition, ownership history, or vehicle specifications.

4. Retrieve the names of cars with a selling price above the overall average selling price and a mileage below the overall average mileage?

```
SELECT
        Name
FROM car_info
        WHERE selling_price > (SELECT
                AVG(selling_price)
            FROM car_info)
            AND mileage < (SELECT
                AVG(mileage)
            FROM car_info);
```

Name
Skoda Slavia 1.0 TSI Ambition
BMW 3 Series Gran Limousine 320Ld Luxury Line
Tata Punch Adventure
Hyundai Creta SX Turbo
Renault Kiger RXT AMT Opt DT
Mahindra XUV700 AX5 Diesel AT
Hyundai Creta SX Diesel AT
Renault Triber RXZ
Hyundai i20 Sportz Diesel
BMW X7 xDrive 30d DPE
Hyundai Creta 1.6 CRDi AT SX Plus
Hyundai Elite i20 Sportz Plus BSIV
Kia Seltos HTX Plus AT D
Kia Seltos HTX Plus AT D
Mahindra Bolero Pik-Up FB 1.7T
Mahindra Scorpio S11 4WD BSIV
Mahindra XUV500 W7
Maruti Vitara Brezza ZXI Plus AT Dual Tone
MG Hector Sharp DCT Dualtone
Tata Harrier XZ Plus
Tata Nexon 1.2 Revotron XZ Plus
Tata Nexon 1.2 Revotron XZA Plus
Toyota Fortuner 2.8 4WD AT BSIV
Toyota Innova Crysta 2.4 G MT 8 STR
Toyota Innova Crysta 2.4 ZX AT
Toyota Innova Crysta 2.7 GX AT 8 STR

Explanation:

This query identifies vehicles that are priced above the market average while offering below-average mileage. Such cars may represent premium or luxury vehicles where customers prioritize brand value, performance, or features over fuel efficiency. This analysis helps understand trade-offs between vehicle price and fuel economy.

5. Calculate the cumulative sum of the selling prices over the years for each car model?

```
SELECT
        Name, year, selling_price,
        SUM(selling_price) OVER (PARTITION BY Name ORDER BY year) AS
cumulative_sum
FROM car_info;
```

Name	year	selling_price	cumulative_sum
Ambassador CLASSIC 1500 DSL AC	2000	75000	75000
Ambassador Classic 2000 DSZ AC PS	1994	99000	99000
Ambassador Grand 1500 DSZ BSIII	2008	122000	122000
Ambassador Grand 2000 DSZ PW CL	2008	200000	200000
Ashok Leyland Stile LE	2013	300000	300000
Audi A3 35 TDI Premium Plus	2017	2600000	2600000
Audi A3 35 TDI Premium Plus	2018	3000000	5600000
Audi A3 40 TFSI Premium	2017	1689999	1689999
Audi A4 1.8 TFSI	2010	730000	730000
Audi A4 2.0 TDI	2014	1500000	3100000
Audi A4 2.0 TDI	2014	1600000	3100000
Audi A4 2.0 TDI 177 Bhp Premium Plus	2013	1200000	1200000
Audi A4 35 TDI Premium Plus	2015	1750000	1750000
Audi A4 35 TDI Premium Plus	2016	2450000	6098999
Audi A4 35 TDI Premium Plus	2016	1898999	6098999
Audi A4 35 TDI Premium Plus	2018	2900000	8998999
Audi A6 2.0 TDI	2014	2200000	2200000
Audi A6 2.0 TDI Design Edition	2013	1689999	1689999
Audi A6 2.0 TDI Premium Plus	2013	2000000	2000000
Audi A6 2.0 TDI Technology	2013	1750000	1750000
Audi A6 35 TFSI Matrix	2019	6223000	18669000
Audi A6 35 TFSI Matrix	2019	5923000	18669000
Audi A6 35 TFSI Matrix	2019	6523000	18669000
Audi Q3 2.0 TDI Quattro Premium Plus	2014	1200000	1200000
Audi Q3 2.0 TDI Quattro Premium Plus	2016	2375000	3575000
Audi Q3 2.0 TDI Quattro Premium Plus	2017	2825000	9225000

Explanation:

This query uses a Window Function to calculate the cumulative selling price for each car model over time. The cumulative total increases year by year and helps analyze long-term pricing trends for individual vehicle models. It provides insight into how the total market value of a model evolves across different manufacturing years.

6. Identify the car models that have a selling price within 10% of the average selling price?

```
SELECT
        Name, selling_price
FROM car_info
    WHERE selling_price BETWEEN (SELECT
                AVG(selling_price)*0.9
        FROM car_info)
    AND
        (SELECT
                AVG(selling_price)*1.1
        FROM car_info);
```

Name	selling_price
Tata Punch Adventure	715000
Hyundai Grand i10 Nios AMT Sportz	675000
Mahindra Bolero Pik-Up FB 1.7T	679000
Maruti Baleno Delta	630000
Maruti Baleno Delta	650000
Maruti Dzire ZXI	700000
Maruti Ignis Zeta AMT	600000
Maruti Ignis Zeta AMT	600000
Maruti Swift AMT VXI	654000
Maruti Swift VXI	600000
Maruti Swift VXI	600000
Datsun GO Plus T VDC	590000
Ford Aspire Titanium BSIV	675000
Ford Aspire Titanium BSIV	675000
Ford Figo 1.2P Titanium MT	675000
Ford Figo Titanium Blu	700000
Ford Freestyle Titanium Plus Petrol...	700000
Ford Freestyle Trend Petrol BSIV	681000
Honda Amaze S CVT Diesel BSIV	700000
Honda Amaze S CVT Diesel BSIV	700000
Honda Amaze S CVT Petrol BSIV	680000
Honda Amaze S i-VTEC	690000
Honda Amaze S i-VTEC	690000
Honda Amaze S i-VTEC	690000
Honda Amaze S i-VTEC	690000
Honda Amaze S i-VTEC	690000

Explanation:
This query identifies vehicles whose selling prices are close to the overall market average. Specifically, it selects cars priced within 10% above or below the average selling price. These vehicles can be considered representative of the market and help identify common pricing benchmarks.

7. Calculate the exponential moving average (EMA) of the selling prices for each car model?

```
SELECT
        Name, year, selling_price,
        AVG(selling_price) OVER
        (PARTITION BY Name ORDER BY year
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)
        AS ema_selling_price
FROM car_info;
```

Name	year	selling_price	ema_selling_price
Ambassador CLASSIC 1500 DSL AC	2000	75000	75000.0000
Ambassador Classic 2000 DSZ AC PS	1994	99000	99000.0000
Ambassador Grand 1500 DSZ BSIII	2008	122000	122000.0000
Ambassador Grand 2000 DSZ PW CL	2008	200000	200000.0000
Ashok Leyland Stile LE	2013	300000	300000.0000
Audi A3 35 TDI Premium Plus	2017	2600000	2600000.0000
Audi A3 35 TDI Premium Plus	2018	3000000	2800000.0000
Audi A3 40 TFSI Premium	2017	1689999	1689999.0000
Audi A4 1.8 TFSI	2010	730000	730000.0000
Audi A4 2.0 TDI	2014	1500000	1500000.0000
Audi A4 2.0 TDI	2014	1600000	1550000.0000
Audi A4 2.0 TDI 177 Bhp Premium Plus	2013	1200000	1200000.0000
Audi A4 35 TDI Premium Plus	2015	1750000	1750000.0000
Audi A4 35 TDI Premium Plus	2016	2450000	2100000.0000
Audi A4 35 TDI Premium Plus	2016	1898999	2032999.6667
Audi A4 35 TDI Premium Plus	2018	2900000	2249749.7500
Audi A6 2.0 TDI	2014	2200000	2200000.0000
Audi A6 2.0 TDI Design Edition	2013	1689999	1689999.0000
Audi A6 2.0 TDI Premium Plus	2013	2000000	2000000.0000
Audi A6 2.0 TDI Technology	2013	1750000	1750000.0000
Audi A6 35 TFSI Matrix	2019	6223000	6223000.0000
Audi A6 35 TFSI Matrix	2019	5923000	6073000.0000
Audi A6 35 TFSI Matrix	2019	6523000	6223000.0000
Audi Q3 2.0 TDI Quattro Premium Plus	2014	1200000	1200000.0000
Audi Q3 2.0 TDI Quattro Premium Plus	2016	2375000	1787500.0000
Audi Q3 2.0 TDI Quattro Premium Plus	2017	2825000	2133333.3333

Explanation:
This query calculates a running average of selling prices for each car model across years. While it uses a cumulative average rather than a true exponential moving average, it helps smooth out price fluctuations and reveals overall pricing trends. Such analysis is useful for understanding long-term market behavior of vehicle models.

8. Identify the car models that have had a decrease in selling price from the previous year?

```
SELECT

        Name, year, selling_price
FROM (
        SELECT

                Name, year, selling_price,
                LAG(selling_price) OVER
                (PARTITION BY Name ORDER BY year)
                AS previous_year_price
        FROM car_info
        )

AS price_changes
        WHERE selling_price < previous_year_price;
```

Name	year	selling_price
Audi A4 35 TDI Premium Plus	2016	1898999
Audi A6 35 TFSI Matrix	2019	5923000
Audi Q7 3.0 TDI Quattro	2009	810000
BMW 5 Series 520d Sport Line	2018	3790000
BMW 6 Series GT 630d Luxury Line	2018	5500000
BMW X1 sDrive20d M Sport	2012	925000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000
BMW X4 M Sport X xDrive20d	2019	5400000

Explanation:
This query uses the LAG() window function to compare a car's selling price with its price from the previous year. It identifies vehicles whose prices have decreased over time, helping measure depreciation trends and understand how vehicle values decline as they age.

9. Retrieve the names of cars with the highest total mileage for each transmission type?

```
WITH TotalMileage AS (
SELECT
    Name, transmission,
    SUM(km_driven) AS total_mileage
FROM car_info
GROUP BY Name, transmission
)
SELECT
    Name, transmission, total_mileage
FROM TotalMileage
WHERE (transmission, total_mileage) IN (
SELECT
    transmission,
MAX(total_mileage)
FROM TotalMileage
GROUP BY transmission
);
```

Name	transmission	total_mileage
Maruti Swift Dzire VDI	Manual	13187291
Toyota Camry 2.5 Hybrid	Automatic	2219848

Explanation:

This query determines which car model has accumulated the highest total distance traveled within each transmission category. By comparing manual and automatic vehicles separately, it identifies the most extensively used vehicles in each segment and provides insight into usage patterns across transmission types.

10. Find the average selling price per year for the top 3 car models with the highest overall selling prices?

```sql
WITH RankedSellingPrices AS (
SELECT

        Name, selling_price,
        RANK() OVER
        (PARTITION BY Name ORDER BY selling_price DESC)
        AS price_rank
FROM car_info
)
SELECT

        Name,year,
        AVG(selling_price) AS avg_selling_price_per_year
FROM car_info
WHERE Name IN (
        SELECT Name
        FROM RankedSellingPrices
        WHERE price_rank <= 3
)
GROUP BY Name, year;
```

Name	year	avg_selling_price_per_year
Hyundai Elite i20 Petrol Asta Option	2018	700000.0000
Hyundai Elite i20 Petrol Sportz	2018	650000.0000
Hyundai Elite i20 Sportz Plus Diesel	2018	750000.0000
Hyundai EON 1.0 Era Plus	2018	340000.0000
Hyundai EON 1.0 Kappa Magna Plus Optional	2018	350000.0000
Hyundai EON Era Plus	2018	299000.0000
Hyundai EON Magna Plus	2018	325000.0000
Hyundai EON Sportz	2018	391333.3333
Hyundai Grand i10 1.2 CRDi Asta	2018	686666.6667
Hyundai Grand i10 1.2 CRDi Sportz	2018	586666.6667
Hyundai Grand i10 1.2 CRDi Sportz Option	2018	550000.0000
Hyundai Grand i10 1.2 Kappa Asta	2018	585000.0000
Hyundai Grand i10 1.2 Kappa Magna AT	2018	555500.0000
Hyundai Grand i10 1.2 Kappa Magna BSIV	2018	495000.0000
Hyundai Grand i10 1.2 Kappa Sportz AT	2018	650000.0000
Hyundai Grand i10 1.2 Kappa Sportz BSIV	2018	541222.2222
Hyundai Grand i10 1.2 Kappa Sportz Dual Tone	2018	570000.0000
Hyundai Grand i10 1.2 Kappa Sportz Option	2018	516000.0000
Hyundai i20 1.2 Asta	2018	704000.0000
Hyundai i20 1.2 Asta Dual Tone	2018	700000.0000
Hyundai i20 1.2 Asta Option	2018	733666.6667
Hyundai i20 1.2 Magna Executive	2018	602500.0000

Explanation:

This query first identifies the top-performing car models based on selling prices using a ranking function. It then calculates the average selling price for those models across different years. The analysis helps track pricing trends of premium vehicles and understand how their market value changes over time.

-- <u>Key Findings and Insights</u> --

Pricing Insights

- BMW models have the highest average selling prices.
- Newer vehicles command significantly higher market value.
- Automatic vehicles are generally priced higher than manual vehicles.

Mileage Insights

- Diesel vehicles offer higher mileage on average.
- Certain compact vehicles dominate fuel efficiency rankings.

Ownership Insights

- First-owner vehicles hold greater resale value.
- Multi-owner vehicles show significant price depreciation.

Market Trends

- Vehicle prices have increased consistently over recent years.
- Automatic transmission adoption is increasing.

Vehicle Pricing Findings

1. Luxury vehicle brands command significantly higher resale values.
2. Newer vehicles consistently achieve higher market prices than older vehicles.
3. Automatic transmission vehicles generally have higher average selling prices than manual vehicles.
4. First-owner vehicles maintain stronger resale value compared to multi-owner vehicles.

Mileage & Performance Findings

1. Diesel vehicles provide higher average mileage compared to petrol vehicles.
2. Vehicles with larger engine capacities generally command higher selling prices.
3. Fuel-efficient vehicles remain highly competitive in the resale market.
4. Certain compact vehicle models dominate mileage rankings.

Market Preference Findings

1. Manual transmission vehicles represent a significant share of the used-car market.
2. First-owner vehicles are the most common ownership category.
3. Petrol and diesel vehicles continue to dominate automotive listings.
4. Popular vehicle models appear repeatedly across different ownership segments.

VALUE & PROFITABILITY FINDINGS

1. Premium vehicle models contribute disproportionately to overall market value.
2. Vehicle age is one of the strongest drivers of depreciation.
3. Ownership history directly impacts resale pricing.
4. High-mileage vehicles often experience lower average market valuation.

KEY ATTRIBUTES :-

Vehicle Information

- Car Name
- Manufacturing Year
- Selling Price

Performance Information

- Mileage
- Engine Capacity
- Maximum Power
- Torque

Vehicle Characteristics

- Fuel Type
- Transmission Type
- Seating Capacity

Ownership Information

- Seller Type
- Ownership Category

Business Value of Dataset :-

This dataset enables the analysis of:

- Vehicle pricing trends across years
- Impact of ownership history on resale value
- Fuel type preferences in the market
- Relationship between mileage and vehicle price
- Performance characteristics of high-value vehicles
- Market demand for automatic versus manual vehicles
- Profitability opportunities within different vehicle segments

To improve the analysis, numerical transformations and data-cleaning techniques were applied to mileage, engine capacity, torque, and power columns before performing SQL-based exploration.

Conclusion :-

This SQL-based analysis of the automotive industry provides valuable insights into vehicle pricing, fuel efficiency, ownership patterns, and overall market trends. By leveraging a dataset containing more than 8,000 vehicle records, the project demonstrates how structured data analysis can reveal the key factors influencing vehicle valuation and consumer preferences.

You can add these concise points to strengthen your conclusion:

- Analyzed **8,000+ vehicle records** to uncover pricing trends and market behavior.
- Identified the **key factors affecting vehicle resale value**, including age, brand, fuel type, transmission, and ownership history.
- Used **SQL aggregate functions** to calculate averages, totals, and category-wise performance.
- Applied **CTEs and subqueries** to simplify complex data analysis and improve query readability.
- Leveraged **window functions** to rank vehicles, compare prices, and identify top-performing categories.
- Discovered that **newer vehicles, first-owner cars, and premium brands** generally command higher resale prices.
- Evaluated the relationship between **fuel efficiency, engine performance, and customer demand**.
- Generated insights that can support **pricing strategies, inventory optimization, and market segmentation**.
- Demonstrated the ability to **transform raw data into actionable business insights** using SQL.
- Showcased practical SQL skills relevant to **Data Analyst, Business Analyst, and Data Science** roles.
- Improved understanding of **consumer purchasing behavior** through data-driven analysis.
- Highlighted the importance of **data analytics in supporting business decision-making** within the automotive industry.